

LAB 1- Part 1

Introduction to IDE and Basic Assembly

Objectives:

WEEK 2

- Introduction to Assembler and Basic Assembly

Grading:

- Total mark = _____/50

Material:

- Computer
- Keil uVision5 IDE

Pre-lab Activity

Objective: Prepare for the in-lab programming tasks.

Before starting the lab, complete the following:

- Read and follow the tutorial:
"Creating an Assembly Project and Run Simulator in Keil – Step-by-Step Tutorial"
(available on Blackboard under *Additional Resources*).
- Make sure you can:
 - Create a new Keil project
 - Add a .s file
 - Build the project
 - Open and use the simulator
 - View register values

LAB Activity 1 – Register Arithmetic [15 points]

Objective: Practice loading registers and performing arithmetic operations.

Task

Write an ARM assembly program that performs the following:

- Load registers with initial values:
 - R2 = 10
 - R3 = 20
 - R4 = 30
- Compute the following expression:

$$R5 = R2 + R3 - R4$$

Requirements

- Use appropriate ARM assembly instructions (e.g., MOV, ADD, SUB)
- Run the program using the **Keil Simulator**
- Single-step through each instruction
- Observe and verify register values (R2–R5)

Reflection:

After single-stepping through your final instruction in Activity 1:

- Insert a screenshot of the register window.
- Circle or highlight registers R2, R3, R4, and R5.
- Explain (in 2–3 sentences) why the final value of R5 makes sense given your expression.

Lab Activity 2 – Negative Numbers and Addition [15 points]

Objective: Work with larger values and signed arithmetic.

Task

Write an ARM assembly program that:

- Loads registers with decimal values:
 - R2 = 500
 - R5 = 100
- Negates both values

- Adds the two negated values together
- Stores the final result in:
 - R7

Requirements

- Demonstrate the use of:
 - Immediate values
 - Two's complement (negation)
 - Arithmetic instructions
- Use the simulator to:
 - Single-step through the code
 - Verify register values after each step

Reflection

After running Activity 2:

- Record the exact hexadecimal value shown in R7.
- Insert a screenshot as evidence.
- Explain why this value represents the correct signed result.

Lab Activity 3 – Code Tracing and Logical Operations [15 points]

Objective: Strengthen understanding of how instructions affect registers.

Given Code

```
LDR R1, =623
MOV R2, #50
AND R1, R1, R2
ORR R3, R1, R2
```

Part A – Manual Interpretation

For each instruction:

- Predict the new value of registers R1, R2, and R3
- Show your work clearly (in binary or hexadecimal where appropriate)

Part B – Simulator Verification

- Enter the same code into Keil
- Run the simulator in **single-step mode**
- Compare simulator results with your manual calculations

Reflection

Before running your code, write down your *predicted* values for R1, R2, and R3 after the final instruction.

After simulation:

- Insert a screenshot of the registers.
- Compare your prediction to the actual values.
- If there was a difference, explain what you misunderstood and how you corrected it.

Post-Lab Questions [40 points]

Note: The answers must be authentic and not AI-generated.

1. What is the Toolchain in the software development flow for microcontrollers?
2. Explain the difference between **assembly** and **linking** during the build process. What are the names of files generated after each step in your project?
3. In your project files, locate one file that comes from CMSIS and one file that comes from Startup. Give the filenames and explain (briefly) what each one does.
4. Describe two common causes of build errors and how you would troubleshoot them in Keil.
5. In Activity 3, why does the AND instruction change the value of R1? Explain conceptually using bitwise logic.
6. What is the difference between MOV R2, #50 and LDR R1, =623 in terms of how data is loaded into registers?
7. What does the memory address of a function (such as __main) represent?
8. Why might an embedded developer need to know the memory locations of functions or variables?
9. What is one concept from this lab that became clearer only after using the simulator (not just reading about it)? Explain using your own experience.
10. If you were teaching this lab to a first-year student, what is one common mistake you would warn them about and why?

Report Submission [5 points]

Submit a report that includes:

- Description of objectives and tasks
- Answer to post-lab questions
- Your full assembly code for each activity
- Screenshots of:
 - Your source code in Keil
 - The simulator showing register values
- Your manual calculations for Activity 3
- A brief comparison between:
 - Your predicted results
 - The simulator results
 - Reports without screenshots or manual work will lose marks.